

Honours Multivariate Analysis

Lecture 4 - Maximum Likelihood Estimation and Testing for Normality

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Course Outline

- ① Introduction, Examples of Multivariate Data [†]
- ② TOOLS
 - ① Visualization and Summary Statistics [†]
 - ② Singular Value Decomposition, Eigenvalue Decomposition and Spectral Decomposition revisited [†]
 - ③ The Multivariate Normal Distribution [†]
 - ④ **Multivariate Maximum Likelihood Estimation** [†]
 - ⑤ Multivariate Inference [†]
- ③ EXPLORATORY ANALYSIS
 - ① Principal Component Analysis [§]
 - ② Factor Analysis [§]
 - ③ Correspondence Analysis [§]
- ④ CONFIRMATORY ANALYSIS
 - ① For grouped Multivariate Data:
 - ① Manova [†]
 - ② Discriminant Analysis [§]
 - ② Regression
 - ① Multivariate Regression [§]
 - ② Canonical Correlation Analysis [§]

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[§]Mr Miguel Rodo

The MVN Density and Likelihood

Consider a random sample, $\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_n$ from the multivariate normal population $\mathbf{X} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, with the $\mathbf{X}_j$'s mutually independent.

The joint density of $\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_n$ is given by

$$\begin{aligned} f(\mathbf{x}_1, \dots, \mathbf{x}_n) &= \prod_{j=1}^n \frac{1}{(2\pi)^{\frac{p}{2}} |\boldsymbol{\Sigma}|^{\frac{1}{2}}} \exp \left[-\frac{1}{2} (\mathbf{x}_j - \boldsymbol{\mu})' \boldsymbol{\Sigma}^{-1} (\mathbf{x}_j - \boldsymbol{\mu}) \right] \\ &= \left(\frac{1}{(2\pi)^{\frac{p}{2}} |\boldsymbol{\Sigma}|^{\frac{1}{2}}} \right)^n \exp \left[-\frac{1}{2} \sum_{j=1}^n (\mathbf{x}_j - \boldsymbol{\mu})' \boldsymbol{\Sigma}^{-1} (\mathbf{x}_j - \boldsymbol{\mu}) \right] \end{aligned}$$

The MVN Density and Likelihood

This joint density, written as a function of the variables, regards the parameters as fixed, albeit unknown, constants.

However, when observations are made, i.e. we are given values for $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n$, then we can consider this expression to be a function of the parameters, referred to as the *likelihood*.

We are therefore trying to ascertain how likely specific values of the parameters $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ (viewed as variable) are, given our fixed observations.

$$\therefore L(\boldsymbol{\mu}, \boldsymbol{\Sigma}) = \left(\frac{1}{(2\pi)^{\frac{p}{2}} |\boldsymbol{\Sigma}|^{\frac{1}{2}}} \right)^n \exp \left[-\frac{1}{2} \sum_{j=1}^n (\mathbf{x}_j - \boldsymbol{\mu})' \boldsymbol{\Sigma}^{-1} (\mathbf{x}_j - \boldsymbol{\mu}) \right]$$

The MVN Density and Likelihood

For ease of notation to follow, define

$$\mathbf{A} = (n - 1)\mathbf{S} = \mathbf{X}'\mathbf{X} - n\bar{\mathbf{x}}\bar{\mathbf{x}}' = \sum_{i=1}^n (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})'$$

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The summation in the exponent of the likelihood function can then be written as

$$\sum_{j=1}^n (\mathbf{x}_j - \boldsymbol{\mu})'\boldsymbol{\Sigma}^{-1}(\mathbf{x}_j - \boldsymbol{\mu}) = \text{tr} [\boldsymbol{\Sigma}^{-1}\mathbf{A}] + n(\bar{\mathbf{x}} - \boldsymbol{\mu})'\boldsymbol{\Sigma}^{-1}(\bar{\mathbf{x}} - \boldsymbol{\mu})$$

Make sure you go through the steps in the notes!

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Make sure you go through the steps in the notes!

We can now write the joint likelihood function as

$$L(\boldsymbol{\mu}, \boldsymbol{\Sigma}) = (2\pi)^{-\frac{np}{2}} |\boldsymbol{\Sigma}|^{-\frac{n}{2}} \exp \left[-\frac{1}{2} \text{tr} [\boldsymbol{\Sigma}^{-1}\mathbf{A}] - \frac{n}{2} (\bar{\mathbf{x}} - \boldsymbol{\mu})'\boldsymbol{\Sigma}^{-1}(\bar{\mathbf{x}} - \boldsymbol{\mu}) \right]$$

MLE of μ

We consider Σ as fixed and maximise

$$\begin{aligned}\log[L(\mu, \Sigma)] &= l(\mu, \Sigma) \\ &= -\frac{np}{2} \log(2\pi) - \frac{n}{2} \log |\Sigma| - \frac{1}{2} \text{tr}[\Sigma^{-1} A] - \frac{n}{2} (\bar{x} - \mu)' \Sigma^{-1} (\bar{x} - \mu)\end{aligned}$$

with respect to μ .

By inspection:

$$\begin{aligned}l &= -\frac{np}{2} \log(2\pi) - \frac{n}{2} \log |\Sigma| - \frac{1}{2} \text{tr}[\Sigma^{-1} A] - \frac{n}{2} (\bar{x} - \mu)' \Sigma^{-1} (\bar{x} - \mu) \\ &\leq -\frac{np}{2} \log(2\pi) - \frac{n}{2} \log |\Sigma| - \frac{1}{2} \text{tr}[\Sigma^{-1} A]\end{aligned}$$

with equality when $\frac{n}{2} (\bar{x} - \mu)' \Sigma^{-1} (\bar{x} - \mu) = 0$, i.e., when

$$\hat{\mu} = \bar{x}.$$

MLE of μ

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$$\begin{aligned}\log[L(\mu, \Sigma)] &= l(\mu, \Sigma) \\ &= -\frac{np}{2} \log(2\pi) - \frac{n}{2} \log |\Sigma| - \frac{1}{2} \text{tr}[\Sigma^{-1} \mathbf{A}] - \frac{n}{2} (\bar{x} - \mu)' \Sigma^{-1} (\bar{x} - \mu)\end{aligned}$$

with respect to μ .

By differentiation:

$$\begin{aligned}\frac{\partial l}{\partial \mu} &= -\frac{n}{2} 2\Sigma^{-1}(\bar{x} - \mu)(-1) && \text{(See Theorem A.1)} \\ n\Sigma^{-1}(\bar{x} - \hat{\mu}) &= \mathbf{0} \\ \hat{\mu} &= \bar{x}\end{aligned}$$

MLE of μ

Showing that $\hat{\mu} = \bar{x}$ is indeed a maximum and not a minimum or saddle point:

$$\frac{\partial^2 l}{\partial \mu \mu'} = \frac{\partial}{\partial \mu'} \{n \Sigma^{-1} \bar{x} - n \Sigma^{-1} \mu\} = -n \Sigma^{-1}$$

which is negative definite. Note that Σ^{-1} is positive definite since Σ is positive definite and symmetric.

MLE of Σ

Let $\hat{\boldsymbol{\mu}} = \bar{\mathbf{x}}$. We first need to find the stationary point of $l(\hat{\boldsymbol{\mu}}, \boldsymbol{\Sigma})$ through differentiation with respect to $\boldsymbol{\Sigma}^{-1}$, then show that the stationary point is a maximum. See Theorems A.2 & A.3.

$$\begin{aligned}l(\hat{\boldsymbol{\mu}} = \bar{\mathbf{x}}, \boldsymbol{\Sigma}) &= -\frac{np}{2} \log(2\pi) - \frac{n}{2} \log |\boldsymbol{\Sigma}| - \frac{1}{2} \text{tr}[\boldsymbol{\Sigma}^{-1} \mathbf{A}] \\&= -\frac{np}{2} \log(2\pi) + \frac{n}{2} \log |\boldsymbol{\Sigma}^{-1}| - \frac{1}{2} \text{tr}[\boldsymbol{\Sigma}^{-1} \mathbf{A}] \\ \frac{\partial l}{\partial \boldsymbol{\Sigma}^{-1}} &= \frac{n}{2} [(\boldsymbol{\Sigma}^{-1})^{-1}]' - \frac{1}{2} \mathbf{A}'\end{aligned}$$

$$\frac{n}{2} \hat{\boldsymbol{\Sigma}} - \frac{1}{2} \mathbf{A} = \mathbf{0}$$

$$n \hat{\boldsymbol{\Sigma}} = \mathbf{A}$$

$$\hat{\boldsymbol{\Sigma}} = \frac{1}{n} \mathbf{A} = \frac{n-1}{n} \mathbf{S}$$

MLE of Σ

To show that this stationary point is a maximum, taking the second order derivative with respect to Σ^{-1} is complicated. Therefore, the likelihood is reparameterised. See the course notes for the gory details.

When substituting $\hat{\mu}$ and $\hat{\Sigma}$ into the likelihood, we end up with the following expression for the maximised likelihood:

$$\begin{aligned} L(\hat{\mu}, \hat{\Sigma}) &= (2\pi)^{-\frac{np}{2}} |\hat{\Sigma}|^{-\frac{n}{2}} \exp \left[-\frac{1}{2} \text{tr} \left[\hat{\Sigma}^{-1} \mathbf{A} \right] - \frac{n}{2} (\bar{x} - \hat{\mu})' \hat{\Sigma}^{-1} (\bar{x} - \hat{\mu}) \right] \\ &= (2\pi)^{-\frac{np}{2}} \left| \frac{1}{n} \mathbf{A} \right|^{-\frac{n}{2}} \exp \left[-\frac{1}{2} \text{tr} \left[n \mathbf{A}^{-1} \mathbf{A} \right] \right] \\ &= (2\pi)^{-\frac{np}{2}} \left| \frac{1}{n} \mathbf{A} \right|^{-\frac{n}{2}} e^{-\frac{np}{2}} \end{aligned}$$

Homework exercise 4.1

Johnson & Wichern exercise 4.18

Find the maximum likelihood estimates of $\mu_{2 \times 1}$ and $\Sigma_{2 \times 2}$ based on the random sample

$$\begin{bmatrix} 3 & 6 \\ 4 & 4 \\ 5 & 7 \\ 4 & 7 \end{bmatrix}$$

from a bivariate normal population.

Sampling Distribution of $\bar{X}$

Reminder of the univariate case:

If $X_1, X_2, \dots, X_n$ are i.i.d random observation from $X \sim N(\mu, \sigma^2)$, then

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

and

$$\frac{(n-1)s^2}{\sigma^2} \sim \chi_{n-1}^2$$

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The multivariate case:

Now suppose we observe $\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_n$, which are i.i.d random vectors drawn from $\mathbf{X} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$.

For the sample mean we have

$$\bar{\mathbf{X}} \sim N_p\left(\boldsymbol{\mu}, \frac{1}{n}\boldsymbol{\Sigma}\right)$$

Sampling Distribution of S

For the sample variance, note that the univariate case can be expressed as

$$\begin{aligned}(n-1)s^2 &\sim \sigma^2 \chi_{n-1}^2 \\ &= \sigma^2 (Z_1^2 + \dots + Z_{n-1}^2) \\ &= (\sigma Z_1)^2 + \dots + (\sigma Z_{n-1})^2\end{aligned}$$

with each $\sigma Z_i = X_i - \mu \sim N(0, \sigma^2)$.

This form is suitably generalized to the basic sampling distribution of the covariance matrix, namely the **Wishart distribution**.

This distribution is the multivariate analogue of the χ^2 -distribution.

Sampling Distribution of S

If we define $\mathbf{Y}_i = \mathbf{X}_i - \boldsymbol{\mu} \sim N_p(\mathbf{0}, \boldsymbol{\Sigma})$ and let $\mathbf{Y}$ represent the $p \times n$ matrix constructed from the n independent observations of $\mathbf{X}_i$, then

$$\mathbf{Y}\mathbf{Y}' = \sum_{i=1}^n (\mathbf{X}_i - \boldsymbol{\mu})(\mathbf{X}_i - \boldsymbol{\mu})' \sim W_p(n, \boldsymbol{\Sigma})$$

We say that $\mathbf{Y}\mathbf{Y}'$ is p -dimensional Wishart distributed with n degrees of freedom.

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We say that $\mathbf{Y}\mathbf{Y}'$ is p -dimensional Wishart distributed with n degrees of freedom.

Similar to the univariate case, when $\bar{\mathbf{X}}$ is substituted for $\boldsymbol{\mu}$, the distribution remains Wishart, but with one less degree of freedom:

$$\sum_{i=1}^n (\mathbf{X}_i - \bar{\mathbf{X}})(\mathbf{X}_i - \bar{\mathbf{X}})' = \mathbf{A} = (n-1)\mathbf{S} \sim W_p(n-1, \boldsymbol{\Sigma})$$

$$\mathbf{S} \sim W_p\left(n-1, \frac{1}{n-1}\boldsymbol{\Sigma}\right)$$

Properties of the Wishart distribution

Given $\mathbf{A} \sim W_p(n, \mathbf{\Sigma})$, we note the following properties:

- The Wishart distribution is a generalization of the χ^2 -distribution, where $W_1(n, \sigma^2) = \sigma^2 \chi_n^2$.

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- If $\mathbf{c}_{p \times 1}$ is a non-zero constant vector, then $\frac{\mathbf{c}' \mathbf{A} \mathbf{c}}{\mathbf{c}' \mathbf{\Sigma} \mathbf{c}} \sim \chi_n^2$.

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- The density only exists if $n > p$.
- The density itself, although not of particular use to us, is given by

$$f(\mathbf{A}) = \frac{|\mathbf{A}|^{\frac{n-p-1}{2}} \exp \left[-\frac{1}{2} \text{tr}(\mathbf{A} \mathbf{\Sigma}^{-1}) \right]}{2^{\frac{p(n-1)}{2}} \pi^{\frac{p(p-1)}{4}} |\mathbf{\Sigma}|^{\frac{n-1}{2}} \prod_{i=1}^p \Gamma \left(\frac{n-i}{2} \right)}$$

for $\mathbf{A}$ positive definite.

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- $\text{tr}(\mathbf{A}) \sim \chi_{np}^2$.

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for $\mathbf{A}$ positive definite.

- $\text{tr}(\mathbf{A}) \sim \chi_{np}^2$.
- $E(\mathbf{A}) = n \mathbf{\Sigma}$.

Summary of Sampling distribution results

Let $\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_n$ be a random sample from $\mathbf{X} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$. Then

- ① $\bar{\mathbf{X}} \sim N_p\left(\boldsymbol{\mu}, \frac{1}{n}\boldsymbol{\Sigma}\right)$
- ② $(n-1)\mathbf{S} \sim W_p(n-1, \boldsymbol{\Sigma})$
- ③ $\bar{\mathbf{X}}$ and $\mathbf{S}$ are independent.
- ④ $\bar{\mathbf{X}}$ and $\mathbf{S}$ are *sufficient statistics* for $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$.

Large Sample behaviour of $\bar{X}$ and S

If $\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_n$ are independent observations from some population with mean $\boldsymbol{\mu}$ and finite, nonsingular covariance $\boldsymbol{\Sigma}$, then

$$\bar{\mathbf{X}} \dot{\sim} N_p\left(\boldsymbol{\mu}, \frac{1}{n}\boldsymbol{\Sigma}\right) \text{ or, equivalently, } \sqrt{n}(\bar{\mathbf{X}} - \boldsymbol{\mu}) \dot{\sim} N_p(\mathbf{0}, \boldsymbol{\Sigma})$$

and

$$n(\bar{\mathbf{X}} - \boldsymbol{\mu})' \mathbf{S}^{-1} (\bar{\mathbf{X}} - \boldsymbol{\mu}) \dot{\sim} \chi_p^2$$

for large $n - p$.

Homework exercise 4.2

Johnson & Wichern exercises 4.19 & 4.21 (combined)

Let $\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_{60}$ be a random sample of size $n = 60$ from an $N_6(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ population. Specify each of the distributions completely (indicate if the distribution is approximate):

① $\bar{\mathbf{X}}$ and $\sqrt{n}(\bar{\mathbf{X}} - \boldsymbol{\mu})$

② $(\mathbf{X}_1 - \boldsymbol{\mu})' \boldsymbol{\Sigma}^{-1} (\mathbf{X}_1 - \boldsymbol{\mu})$

③ $(n - 1)\mathbf{S}$

④ $n(\bar{\mathbf{X}} - \boldsymbol{\mu})' \boldsymbol{\Sigma}^{-1} (\bar{\mathbf{X}} - \boldsymbol{\mu})$

⑤ $n(\bar{\mathbf{X}} - \boldsymbol{\mu})' \mathbf{S}^{-1} (\bar{\mathbf{X}} - \boldsymbol{\mu})$

Testing the Assumption of Normality

Up until now we have been **assuming** MVN observations.

To determine whether the observations $\mathbf{X}_j$ appear to violate the assumption that they jointly came from a multivariate normal population, we address the following questions:

- 1 Do the marginal distributions of the elements of $\mathbf{X}$ appear to be normal?
- 2 Do the scatter plots of the pairs of observations on different characteristics give the elliptical appearance expected from normal populations?
- 3 Are there any “wild” observations that should be checked?

Univariate assessment – QQ-plots

QQ-plots, or quantile-quantile plots:

- Quick, visual way of assessing how closely the distribution of observed data matches some theoretical distribution
- Plot the sample quantile versus the quantile one would expect to observe if the observations were actually normally distributed

Univariate assessment – QQ-plots

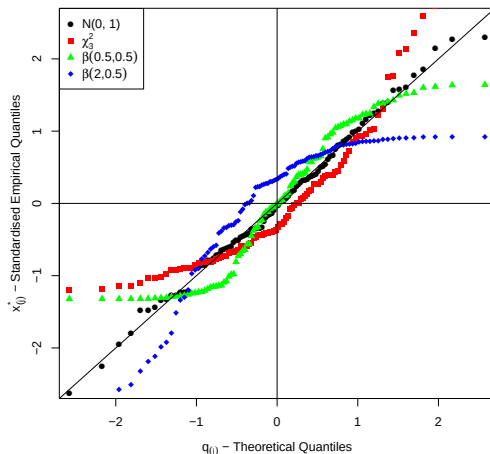
QQ-plots, or quantile-quantile plots:

- If we order the n observations such that $x_{(1)} \leq x_{(2)} \leq \dots \leq x_{(n)}$, then if the $x_{(j)}$ are distinct, exactly j observations will be less than or equal to $x_{(j)}$
- The proportion of the sample at or to the left of $x_{(j)}$, i.e. $\frac{j}{n}$, is often approximated by $\frac{j - \frac{1}{2}}{n}$ for analytical convenience
- The quantiles for a standard normal distribution are defined as those values $q_{(j)}$ such that

$$\Pr[Z \leq q_{(j)}] = \int_{-\infty}^{q_{(j)}} \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} dz = p_{(j)} = \frac{j - \frac{1}{2}}{n}$$

Univariate assessment – QQ-plots

Under the assumption of normality, a plot of $(q_{(j)}, x_{(j)}^*)$ for all j should be a straight line through the origin, where $x_{(j)}^*$ denotes the standardised values of x



Univariate assessment – correlation test

To formally test the linearity, we can calculate the correlation coefficient for the QQ plot, defined as

$$r_q = \frac{\sum_{j=1}^n (x_{(j)} - \bar{x})(q_{(j)} - \bar{q})}{\sqrt{\sum_{j=1}^n (x_{(j)} - \bar{x})^2} \sqrt{\sum_{j=1}^n (q_{(j)} - \bar{q})^2}}$$

and compare it to a table of critical values, given in Appendix B.

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and compare it to a table of critical values, given in Appendix B.

Many other formal tests for univariate normality exist, for example:

- Shapiro-Wilk
- Anderson-Darling
- Jarque-Bera
- Lilliefors test
- etc.

Multivariate assessment

We can now compare the contours of constant density from observed data with the ellipsoid as defined in chapter 3.3, where a set of bivariate outcomes $\mathbf{x}$ such that

$$(\mathbf{x} - \boldsymbol{\mu})' \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}) \leq \chi_p^2(\alpha)$$

has probability $1 - \alpha$.

- Typically we calculate the above for $\alpha = 0.5$ and calculate the proportion of points for which the squared distance is less than $\chi_p^2(0.5)$.
- If this deviates from 50%, it is evidence against the assumption of normality.

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Again, many other formal test can be applied:

- Mardia
- Henze-Zirkler
- Royston
- etc.

Multivariate assessment

We can also construct a chi-square plot based on the assumption that, given underlying normality, the squared distances, $d_j^2 = (\mathbf{x}_j - \bar{\mathbf{x}})' \mathbf{S}^{-1} (\mathbf{x}_j - \bar{\mathbf{x}})$ should behave like chi-square random variables.

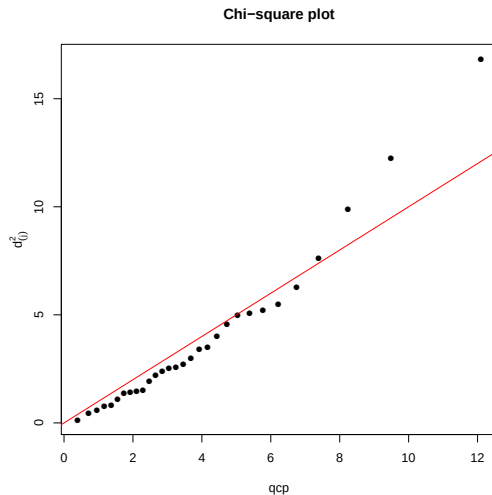
To construct these plots,

- 1 Calculate $d_j^2 = (\mathbf{x}_j - \bar{\mathbf{x}})' \mathbf{S}^{-1} (\mathbf{x}_j - \bar{\mathbf{x}})$.
- 2 Order the squared distances from smallest to largest as $d_{(1)}^2 \leq d_{(2)}^2 \leq \dots \leq d_{(n)}^2$.
- 3 Graph the pairs $\left(q_{c,p} \left(\frac{j - \frac{1}{2}}{n} \right), d_{(j)}^2 \right)$ where $q_{c,p} \left(\frac{j - \frac{1}{2}}{n} \right) = \chi_p^2 \left(\frac{n - j + \frac{1}{2}}{n} \right)$.

If the variables are multivariate normal distributed, the plot should be a straight line through the origin.

Example

Example 4.14 on page 186 of Johnson & Wichern. Solution in `Lecture4.R`.



Caution!

It is important to note the following crucial drawbacks of all measures of fit:

With small samples, only severe deviations will indicate lack of fit, whilst very large samples will invariably produce statistically significant lack of fit.

Homework exercise 4.3

Johnson & Wichern exercises 4.28 & 4.29 (combined)

Consider the air pollution data

- 1 Construct a QQ-plot for the solar radiation measurements and carry out a test for normality based on the correlation coefficient r_q . You are not prescribed to use a specific α -value; what can you report based on Table B.1?

Now examine the pairs $X_5 = \text{NO}_2$ and $X_6 = \text{O}_3$ for bivariate normality.

- 2 Calculate the distances $(\mathbf{x}_j - \bar{\mathbf{x}})' \mathbf{S}^{-1} (\mathbf{x}_j - \bar{\mathbf{x}})$, $j = 1, 2, \dots, 42$, where $\mathbf{x}'_j = [x_{j5}, x_{j6}]$.
- 3 Determine the proportion of observations $\mathbf{x}'_j = [x_{j5}, x_{j6}]$, $j = 1, 2, \dots, 42$ falling within the approximate 50% probability contour of a bivariate normal distribution.
- 4 Construct a chi-square plot for the ordered distances in Part 3.